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Altitude aware trajectory prediction methods for non towered terminal airspace

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In non-towered terminal airspace, the lack of real-time air traffic control makes accurate trajectory prediction a critical task for aviation safety. To address the challenges of trajectory prediction caused by frequent altitude changes in this airspace, we propose ACTrajNet, an altitude-aware trajectory prediction model. The model independently extracts altitude features using temporal convolutional networks(TCN), it then incorporates a channel attention fusion mechanism to dynamically fuse altitude features into the trajectory representation across different channels. This significantly enhances the model's ability to capture complex vertical flight patterns. ACTrajNet outperforms state-of-the-art models on the TrajAir dataset, achieving up to 2.47% and 2.42% improvement in ADE(Average Displacement Error) and FDE(Final Displacement Error), respectively, compared to the baseline model TrajAirNet. These results demonstrate the potential of ACTrajNet to improve trajectory prediction under complex flight conditions.

Globally many airports are not equipped with control towers for air traffic management. According to statistics from the Federal Aviation Administration (FAA), only 4% of the approximately 20,000 registered airports in the U.S. are equipped with control towers[1], while the accident rate at non-towered airports is 36 times higher than that at towered airports[2],77% of non-fatal incidents occur in terminal airspace[3]. Due to the absence of a unified air traffic management system, the lack of timely information sharing between aircraft increases the risk of accidents. As a result, pilots need to rely on accurate trajectory prediction technologies to anticipate the movements of other aircraft, thereby reducing the potential for accidents.

In non-towered terminal airspace, aircraft missions are complex and dynamic. Their flight trajectories are often adjusted frequently based on mission requirements, pilot decisions, and external environmental factors, with vertical altitude fluctuations being particularly significant[4]. Fig. 1(a) shows a portion of flight trajectories from the Amerlia42-Mini dataset, representing towered terminal airspace, while Fig. 1(b) presents the distribution characteristics of trajectories from the Trajairnet dataset, representing non-towered terminal airspace. In comparison, the latter exhibits greater dynamic fluctuations in the altitude dimension. However, existing trajectory prediction methods exhibit significant shortcomings in extracting altitude features. Traditional methods, constrained by the assumption of steady motion, struggle to capture the nonlinear dynamics of rapid altitude changes. Some deep learning-based approaches have partially extracted altitude features, but they generally employ static fusion strategies, failing to dynamically adjust the importance of altitude features based on actual fluctuation intensity. This limitation reduces the model's sensitivity to altitude variations in complex scenarios, ultimately affecting prediction accuracy. Therefore, there is an urgent need for a trajectory prediction method. This method should effectively extract altitude features and dynamically integrate multi-dimensional information based on real fluctuation intensity to improve prediction accuracy in complex scenarios.

This paper proposes ACTrajNet, an altitude-aware trajectory prediction model. For the first time, we introduce the independent extraction of altitude features and their dynamic fusion via channel attention, enabling the model to adaptively focus on key channels based on altitude fluctuations. This allows for precise modeling of trajectories with frequent altitude changes. The proposed method effectively leverages altitude characteristics, improving the accuracy of trajectory prediction.

In summary, our contributions are as follows:

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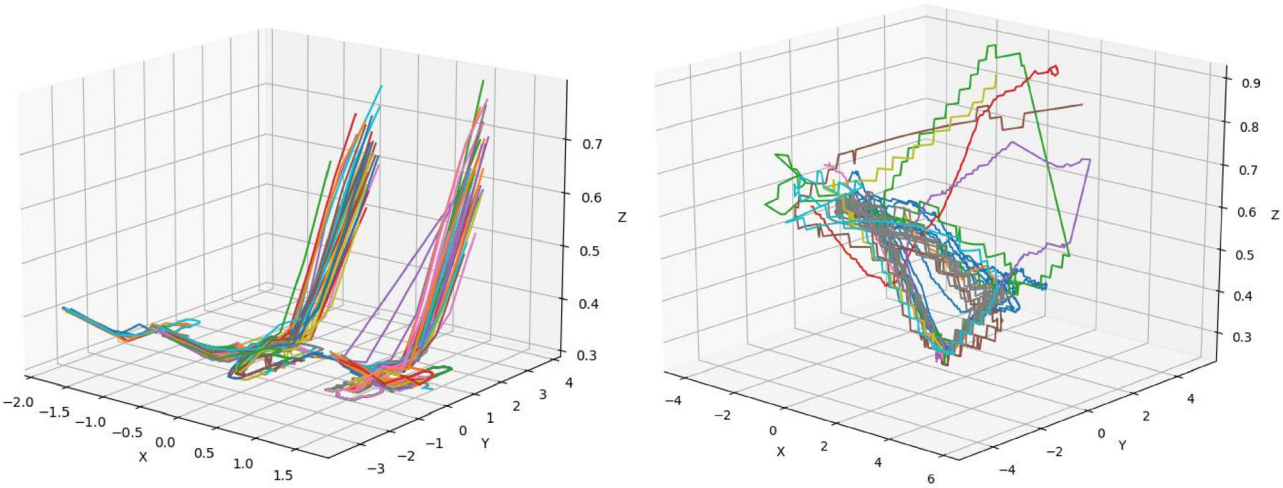


(a) Amerlia42-Mini (b) TrajAir

Fig. 1. Visualization of aircraft trajectory datasets in towered and non-towered terminal airspace. Different colors represent different aircraft. (**a**) shows a portion of trajectory data from Amerlia42-Mini, which represents towered terminal airspace; (**b**) shows a portion of trajectory data from TrajAir, which represents non-towered terminal airspace. Both datasets have the same trajectory sampling interval of one frame.

- We propose the altitude-aware trajectory prediction model ACTrajNet, which uses TCN to independently extract altitude features and dynamically allocate weights through channel attention, enhancing the model's ability to represent trajectories with frequent vertical fluctuations and improving prediction performance.
- We introduce three altitude feature fusion strategies and experimentally validate that the channel attention fusion method is the most effective.
- This paper is the first to propose the independent extraction of altitude features and dynamic fusion through channel attention, achieving accurate modeling of trajectories with frequent vertical fluctuations.

Related works

Aircraft trajectory prediction research falls into three main categories: traditional methods, machine learning, and deep learning, with altitude-aware approaches increasingly enhancing prediction performance.

Traditional methods

Traditional methods fall into two categories: dynamic models and state estimation theory. Dynamic models integrate flight dynamics, aircraft kinematics, and external factors to predict positions by solving differential or difference equations. Thipphavong et al.[5] proposed adaptive model weight adjustment for better climb-phase accuracy, while Gardi et al.[6] developed a dynamic model based on the Base of Aircraft Data (BADA). However, these methods struggle in non-towered terminal airspace, where high flexibility and rapid changes hinder manual feature extraction.

State estimation methods predict trajectories by modeling aircraft motion based on current states. Single-model approaches, like the Kalman filter[7] and its variants[8], use fixed system models, while multi-model estimation (IMM)[9] handles multiple flight modes more effectively. IMM enhances accuracy with diverse dynamic models, improving predictions for complex trajectories[10–12]. However, these methods assume smooth flight paths and struggle with frequent altitude changes and complex takeoff/landing tasks, limiting their applicability in non-towered terminal airspace.

Machine learning-based methods

Machine learning-based methods outperform dynamic models by automatically learning complex patterns from historical data, making them ideal for large-scale and complex systems. Hamed et al.[13] showed that linear regression generalizes better than point mass models, while Tastambekov et al.[14] applied local linear regression for short- to medium-term predictions. Barratt et al.[15] introduced a trajectory probability generation model for terminal areas, and Zhao et al.[16] leveraged clustering and bias correction for commercial aviation.

However, despite capturing complex patterns, these methods lack the adaptability and real-time precision of state estimation approaches, making them unsuitable for non-towered terminal airspace.

Deep learning-based methods

Machine learning-based methods struggle with the dynamic complexity of non-towered terminal airspace, making deep learning the preferred approach.

Hao et al.[17] proposed a CNN-LSTM method. The method employs a CNN integrated with social pooling and attention mechanisms to extract the spatial features of fighter jets. An LSTM is then employed to capture long-term dependencies. However, this method mainly emphasizes horizontal information and shows insufficient

sensitivity to altitude changes. Sun et al.[18] employed a spatiotemporal graph attention network (ST-GAT). They used a Transformer to extract temporal features and a Graph Attention Network (GAT) for spatial features. Yet, the altitude dimension is given little attention, leading to decreased prediction accuracy. Fan et al.[19] introduced the GLAR-GCN, which considers both global and local attribute relationships. However, altitude information is not fully integrated when constructing the attribute graph. Luo et al.[20] developed a long-term prediction model based on a Transformer. This model uses self-attention to extract time-series features and incorporates a trajectory stabilization module. Nevertheless, due to minimal altitude variations, altitude changes are easily overlooked. Zhang et al.[21] utilized wavelet transform for time-frequency analysis to capture both global flight trends and local details. However, the wavelet decomposition process may diminish sensitivity to altitude changes. Guo et al.[22] proposed FlightBERT++ and adopted a non-autoregressive multi-step prediction to avoid error accumulation. However, its encoder relies solely on conventional spatiotemporal patterns and lacks a dedicated module for extracting altitude dynamics. Yang et al.[23] used historical ADS-B data and a Bi-LSTM for short-term prediction. A single sequence memory unit, however, struggles to effectively capture altitude features. Ma et al.[24] proposed the Attention-TCN-GRU, which mixes altitude data with horizontal motion features during encoding. This approach results in insufficient capture of vertical dynamics. Tong et al.[25] constructed a Transformer-based trajectory prediction model. Due to the relatively stable altitude variations and weak temporal dependencies, the model's attention distribution may be biased toward latitude and longitude, resulting in insufficient capture of altitude features. Xu et al.[26] proposed the Bayesian Ensemble Graph Attention Network (BEGAN), which divides the large airspace into overlapping grids for independent training and incorporates a spatiotemporal graph attention mechanism, offering a new approach to trajectory prediction with frequent altitude changes in terminal areas. Subsequent research, PIGAT[27], further introduced partial differential equations based on fluid queuing theory to enhance prediction reliability; however, its modeling of vertical motion remains insufficient.

Thus, there is still a need for models that specifically address frequent altitude fluctuations in non-towered terminal airspace to further enhance prediction accuracy in complex flight scenarios.

Altitude-aware based methods

Altitude awareness in trajectory prediction emphasizes altitude features to enhance accuracy, addressing the shortfall of traditional methods that weaken vertical dynamics. Thipphavong et al.[5] proposed an adaptive model weight adjustment method, which improves prediction accuracy by capturing the temporal characteristics of altitude changes based on aircraft kinematics. However, this approach does not address the generalization issue for nonlinear ascent and descent scenarios. Shi et al.[28] proposed a constrained LSTM model that utilizes climb peak constraints during the ascent phase. During the descent phase, it applies runway direction constraints. This approach aims to fully exploit altitude change characteristics, but it relies on manual physical constraints, limiting its flexibility. Shi et al.[29] proposed the LSTM-ARIMA model, using LSTM as the primary prediction model to forecast three-dimensional information, followed by a secondary model that separately models altitude using ARIMA. However, the two-stage design leads to high computational complexity.

In contrast, our proposed ACTrajNet automatically extracts and adaptively fuses altitude features, eliminating manual intervention and computational redundancy, thereby further improving prediction performance.

Methodology

Problem formulation

The model we use is based on a probabilistic generative approach, so we define the trajectory prediction problem as forecasting the probability distribution of all aircraft in a given scenario, based on the historical trajectories of the aircraft and environmental information. Let $X_t^a = (x_i^a, y_i^a, z_i^a)$ and ϕ_t^a represent the position information of aircraft *a* and the environmental information at time t,respectively. Typically, environmental information can include terrain, maps, weather, and wind information. However, in our work, we only consider the wind factor in the environmental information. $X_{t_1:t_2}^{1:A}$ and $\phi_{t_1:t_2}^{1:A}$ represent all trajectories and environmental information for all aircraft $\{1, \ldots, A\}$ in the scenario within the observation period $\{t_1, \ldots, t_2\}$. The prediction task is to estimate the distribution of future trajectories $\hat{X}_{t_{\text{obs}}:t_{\text{pred}}+t_{\text{obs}}}^{1:A}$, given the historical trajectories $X_{1:t_{\text{obs}}}^{1:A}$ and environmental information $\phi_{1:t_{\text{obs}}}^{1:A}$, the calculation formula is Eq. (1):

$$\hat{X}_{t_{\text{obs}}:t_{\text{pred}}+t_{\text{obs}}}^{1:A} \sim p\left(\hat{X}_{t_{\text{obs}}:t_{\text{pred}}+t_{\text{obs}}}^{1:A} \mid X_{1:t_{\text{obs}}}^{1:A}, \phi_{1:t_{\text{obs}}}^{1:A}\right) \tag{1}$$

where t_{obs} is the observation window and t_{pred} is the prediction window.

ACTrajNet architecture

ACTrajNet leverages a channel attention fusion mechanism that dynamically assigns weights to different trajectory channels based on historical altitude changes. This fused representation, combined with environmental information, is then passed to a CVAE, which learns high-level trajectory features in a supervised manner to make precise predictions.

As shown in Fig. 2, ACTrajNet follows an encoder-decoder structure, where the encoder consists of a TCN and a CNN for feature extraction of trajectory data and environmental information, respectively. Meanwhile, the altitude features encoded by the TCN are further processed through a channel attention fusion mechanism to dynamically weight and fuse the historical trajectory data. The CVAE then uses historical trajectory encodings and environmental data as inputs, with future trajectory features as supervisory conditions, to learn the spatial distribution and predict future positions using the Verlet integration algorithm.

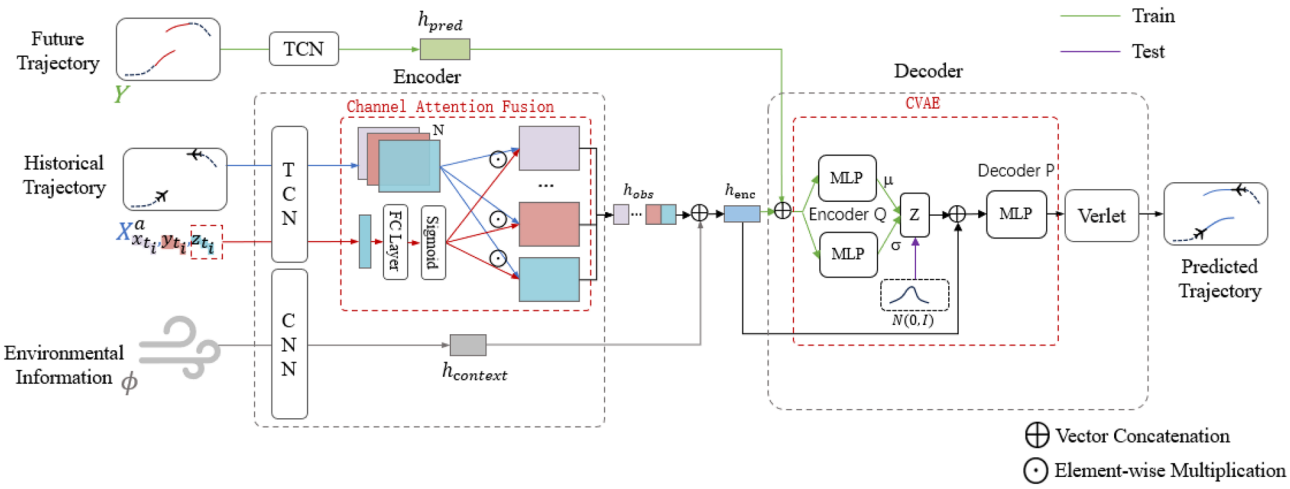


Fig. 2. ACTrajNet model architecture. The green line represents the training phase, and the purple line represents the testing phase.

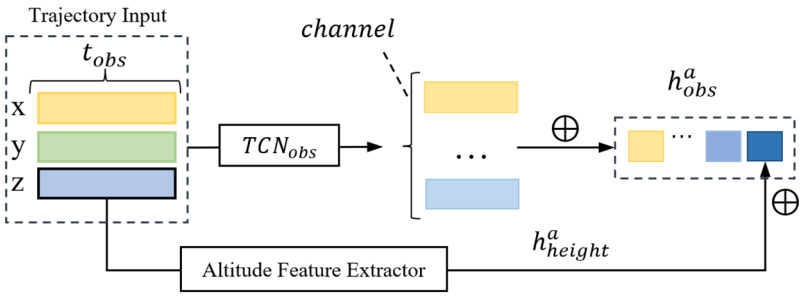


Fig. 3. Vertical channel concatenation.

Encoder module

Input feature encoding

Compared to recurrent models like LSTM and GRU, TCN have a simpler structure and proven effectiveness for time-series tasks[30]. Thus, we use TCN to extract 3D position features from trajectory data, as shown in Eqs. (2) and (3):

$$h'^a_{obs} = \mathrm{TCN}_{\mathrm{obs}}(X^a_{1:t_{obs}}), a \in \{1, ..., A\} \tag{2}$$

$$h^a_{pred} = \mathrm{TCN}_{\mathrm{pred}}\left(X^a_{t_{obs}:t_{obs}+t_{pred}}\right) \tag{3}$$

Where h'^a_{obs} and h^a_{pred} denote the hidden state representations of aircraft a historical and future 3D trajectories respectively. $\mathrm{TCN}_{\mathrm{obs}}$ and $\mathrm{TCN}_{\mathrm{pred}}$ are TCN network modules with different parameters, used for encode the historical trajectory features and future trajectory features, respectively.

To capture wind characteristics, we focus on its horizontal effects–along and perpendicular to the runway. To fully leverage the spatial distribution characteristics of wind data, we employed CNN to capture local spatial patterns The wind encoding is shown in Eq. (4).

$$h^a_{\mathrm{context}} = \mathrm{CNN}(\phi^a_{1:t_{\mathrm{obs}}}),\ a \in \{1, \ldots, A\} \tag{4}$$

Altitude feature fusion module

Given the rich vertical flight patterns in non-towered terminal airspace, we propose three altitude feature fusion methods–Vertical Channel Concatenation (VCC), Horizontal Channel Concatenation (HCC), and Channel Attention Fusion (CAF). It is important to note that these three fusion mechanisms are mutually exclusive, and only one can be selected for use in the experiments.

VCC directly concatenates the extracted altitude representation h^a_{height} to the flattened high-dimensional trajectory representation h'^a_{obs}. The structure of the VCC is shown in Fig. 3. The calculation formulas are Eqs. (5) and (6).

$$h^a_{\mathrm{height}} = \phi(z^a_{1:t_{\mathrm{obs}}}) \tag{5}$$

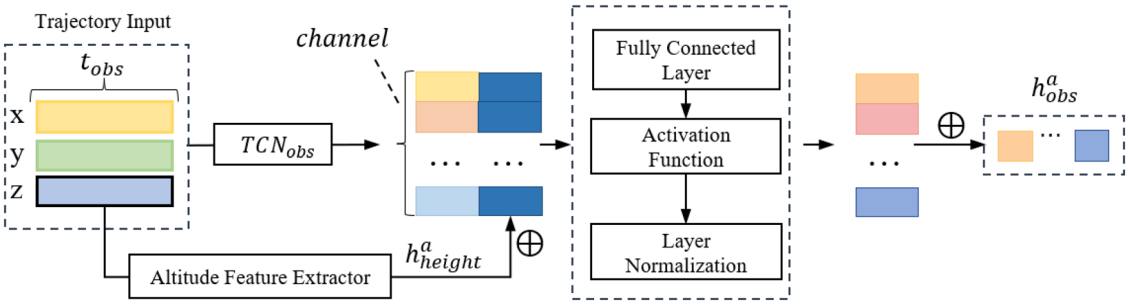


Fig. 4. Horizontal channel concatenation.

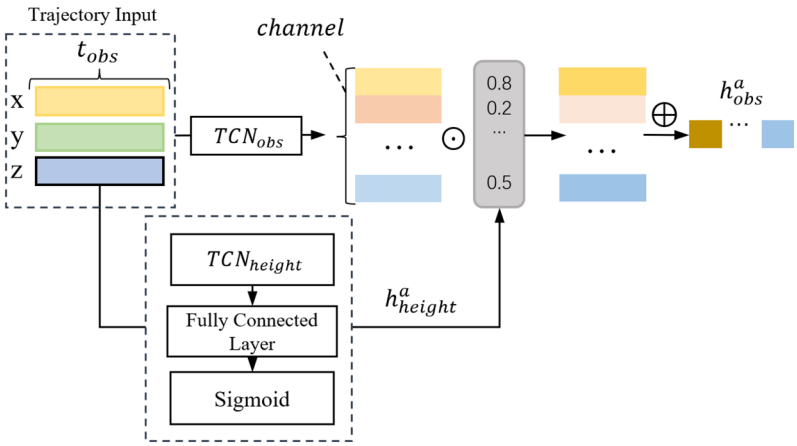


Fig. 5. Channel attention fusion.

$$h^a_{\text{obs}} = \text{Flatten}(h'^a_{\text{obs}}) \oplus h^a_{\text{height}} \tag{6}$$

where $z^a_{1:t_{\text{obs}}}$ denotes the altitude data for aircraft a, with $a \in \{1, \ldots, A\}$, h^a_{height} is its hidden altitude representation, and $\phi(\cdot)$ is the feature extraction function for altitude, corresponding to the altitude feature extractor in Fig. 3, $\text{Flatten}(\cdot)$ denotes flattening, and $\oplus$ indicates concatenation.

HCC concatenates the extracted altitude representation h^a_{height} to the end of each channel of the high-dimensional trajectory representation h'^a_{obs}, resulting in h'^{a*}_{obs}. The structure of the HCC is shown in Fig. 4.The calculation formulas are Eqs. (7) and (8):

$$h^a_{\text{height}} = \phi\left(z^a_{1:t_{\text{obs}}}\right) \tag{7}$$

$$h'^{a*}_{\text{obs}} = h'^a_{\text{obs}_i} \oplus h^a_{\text{height}} \tag{8}$$

where $z^a_{1:t_{\text{obs}}}$ is the altitude data for aircraft a, with $a \in \{1, \ldots, A\}$, $\phi(\cdot)$ is the altitude feature extraction function, corresponding to the altitude feature extractor in Fig. 4, and $h'^a_{\text{obs}_i}$ represents the i-th channel vector in the high-dimensional representation space of the trajectory for aircraft a.

This method enhances altitude feature contributions in each channel. Next, a fully connected layer (FCL) then adjusts the channel scale and fuses the information, followed by ReLU activation and layer normalization. Finally, the newly flattened trajectory representation h^a_{obs} is input into the subsequent model. This process can be expressed by Eq. (9):

$$h^a_{\text{obs}} = \text{Flatten}(\text{Norm}(\text{ReLU}(h'^{a*}_{\text{obs}} W_{\text{height}} + b_{\text{height}}))) \tag{9}$$

where $\text{Flatten}(\cdot)$ flattens the data, $\text{Norm}(\cdot)$ performs layer normalization, $\text{ReLU}(\cdot)$ is the activation function, W_{height} is the learnable weight matrix, and b_{height} is the learnable bias vector.

CAF is the chosen fusion mechanism for ACTrajNet. It plays a crucial role in the model, as illustrated in Fig. 5, which shows its detailed structure.

The extracted altitude feature h^a_{X} is passed through a FCL to perform dimensionality transformation, mapping it to the same channel dimension as h'^a_{obs}. Then, the Sigmoid function is applied to obtain the attention weights for each channel in the high-dimensional representation space of the trajectory. These weights are then multiplied with the trajectory high-dimensional representation on a per-channel basis to obtain the

new trajectory representation h^a_{obs}. Finally, a concatenation operation is performed on the newly constructed trajectory representation channel data to produce the output of the CAF. The calculation formulas are Eqs. (10) and (11):

$$h^a_{\text{height}} = \text{TCN}_{\text{height}}\left(z^a_{1:t_{\text{obs}}}\right) \tag{10}$$

$$h^a_{\text{obs}} = \text{Flatten}\left(\text{sigmoid}\left(h^a_{\text{height}} W_{\text{height}} + b_{\text{height}}\right) \odot h'^a_{\text{obs}}\right) \tag{11}$$

where $z^a_{1:t_{\text{obs}}}$ is the altitude data for aircraft a, with $a \in \{1, \ldots, A\}$, $\text{TCN}_{\text{height}}$ extracts altitude features. $\text{Flatten}(\cdot)$ flattens the input, $\text{sigmoid}(\cdot)$ is the activation function, W_{height} and b_{height} are learnable parameters, and $\odot$ represents the per-channel multiplication.

Decoder module

The decoder module consists mainly of two parts: CVAE and Verlet integration. The CVAE learns the trajectory distribution and generates the Verlet acceleration parameters, which are then used by Verlet integration to derive the predicted trajectory.

Before inputting into the decoder, the historical trajectory data and environmental information are integrated into h^a_{enc}, and this process can be expressed by Eq. (12):

$$h^a_{\text{enc}} = h^a_{\text{obs}} \oplus h^a_{\text{context}} \tag{12}$$

The CVAE uses an encoder-decoder architecture with MLP. In the training phase, the encoder takes the integrated representation vector h^a_{enc} (containing historical trajectory and environmental data) and the real future trajectory h^a_{pred}. The task of the encoder in the CVAE is to map the inputs to a latent Gaussian distribution $Q(z \mid h^a_{\text{pred}}, h^a_{\text{enc}})$. This process can be expressed by Eq. (13):

$$z \sim Q(z \mid h^a_{\text{pred}}, h^a_{\text{enc}}) \quad for\ train \tag{13}$$

where $\text{Q}(\cdot)$ represents the encoder in the CVAE.

In the testing phase, the latent variable z is sampled from a standard normal prior distribution, This process can be expressed by Eq. (14):

$$z \sim N(0, I) \quad for\ test \tag{14}$$

where $z \sim N(0, I)$ denotes a multivariate Gaussian distribution with zero mean and identity covariance matrix.

In the CVAE, the decoder uses z and the historical input encoding h^a_{enc} as conditional information to reconstruct the representation of the future trajectory. This process can be expressed by Eq. (15):

$$h^a_{\text{cvae}} \sim P(h^a_{\text{cvae}} \mid z, h^a_{\text{enc}}) \tag{15}$$

To incorporate the aircraft motion state, we adopts a general dynamics formula[31], which is in the form of forward Verlet integration that provides a constant velocity model when there is zero acceleration. The CVAE does not directly predict the positions of the trajectory but instead predicts the Verlet acceleration, which is then used to compute the absolute position of each aircraft. This method captures both the trajectory's probability distribution and the physical dynamics, leading to improved prediction accuracy. The CVAE output is passed through an MLP to produce the appropriately dimensioned Verlet acceleration tensor, the calculation formula is Eq. (16):

$$s^a_{t_{\text{obs}}:t_{\text{obs}}+t_{\text{pred}}} = \text{MLP}(h^a_{\text{cvae}}) \tag{16}$$

where $\text{MLP}(\cdot)$ is a multilayer perceptron.

Based on the Verlet acceleration tensor, the Verlet integration is used to compute the predicted trajectory coordinates, which can be expressed by Eq. (17):

$$x^{1:A}_{t+1} = 2x^{1:A}_t - x^{1:A}_{t-1} + s^{1:A}_t \Delta t^2 \tag{17}$$

Loss function

The training of the model involves two optimization objectives: the first is the trajectory prediction loss L_{traj}, which measures the difference between the predicted trajectory and the real trajectory(ground-truth); the second is the CVAE loss L_{cvae}, which measures the difference between the latent distribution learned by the encoder and the prior distribution (predefined as a standard normal distribution).

L_{traj} is measured by calculating the Mean Squared Error (MSE) between the predicted trajectory and the real trajectory. The goal of this part is to directly optimize the model accuracy in predicting future trajectories. The mathematical expression of the trajectory prediction loss is shown in Eq. (18):

$$L_{\text{traj}} = \text{MSE}\left(X^a_{t_{\text{obs}}:t_{\text{obs}}+t_{\text{pred}}}, \hat{X}^a_{t_{\text{obs}}+t_{\text{pred}}}\right) \tag{18}$$

L_{cvae} consists of two parts: one is the reconstruction loss (which is also included in the trajectory prediction loss), and the other is the Kullback-Leibler (KL) divergence loss, which measures the difference between the latent space distribution learned by the model and the predefined standard normal distribution. The goal of the

KL divergence loss is to prevent the model from learning excessive details and noise by constraining the latent space distribution to be close to the prior distribution. This improves the ability of the model to generalize and helps avoid overfitting. The mathematical expression for the CVAE loss is given by Eq. (19):

$$L_{\text{cvae}} = D_{\text{KL}}\left(Q(z \mid h^{a}_{\text{pred}}, h^{a}_{\text{enc}}) \parallel N(0, I)\right) \tag{19}$$

where $Q(z \mid h^{a}_{\text{pred}}, h^{a}_{\text{enc}})$ is the latent space distribution learned by the CVAE encoder, $N(0, I)$ is the standard normal distribution, and $D_{\text{KL}}(\cdot)$ represents the KL divergence.

The total loss of the ACTrajNet model is the weighted sum of L_{traj} and L_{cvae}, as shown in Eq. (20):

$$L_{\text{total}} = L_{\text{traj}} + L_{\text{cvae}} \tag{20}$$

During training, by minimizing these two errors, the model learns how to generate diverse future trajectories that align with the real-world data, given the condition of past trajectories.

Experiments

Dataset

The TrajAir dataset[2] we used was created by the team at Carnegie Mellon University and collected from the Pittsburgh-Butler regional airport in Pennsylvania, USA. The data was obtained via ADS-B receivers. The airport is a non-towered, single-runway general aviation facility serving various aircraft. Aircraft follow a left-hand traffic pattern during departure and arrival, maintaining a counterclockwise turn for climb or descent. Fig. 6 shows a visualization of the airport trajectory data, where the red curves represent the flight paths, with the color gradient indicating the pressure altitude of the aircraft, ranging from low (light) to high (dark). The flight direction and altitude changes reflect the flight patterns and complexity of different aircraft, particularly during climbing and descending turns, where the relationship between vertical speed changes and horizontal displacement speed is especially notable.

The dataset includes four 7-days subsets. Each dataset is divided into samples based on scenarios, with each sample containing the trajectory of at least one aircraft. A scenario starts when at least one aircraft is detected as active or enters a critical threshold and ends when all aircraft leave the terminal airspace or become inactive. The sizes of the individual subsets are shown in Table. 1.

Evaluation metrics

We introduce three evaluation metrics for assessing the accuracy of trajectory prediction models: Average Displacement Error (ADE), Final Displacement Error (FDE), and Maximum Displacement Error (MDE). These metrics quantify the discrepancy between predicted and ground-truth trajectories from different perspectives. All metrics are measured in kilometers (km) and are used to quantify the spatial distance between predicted and ground-truth points.

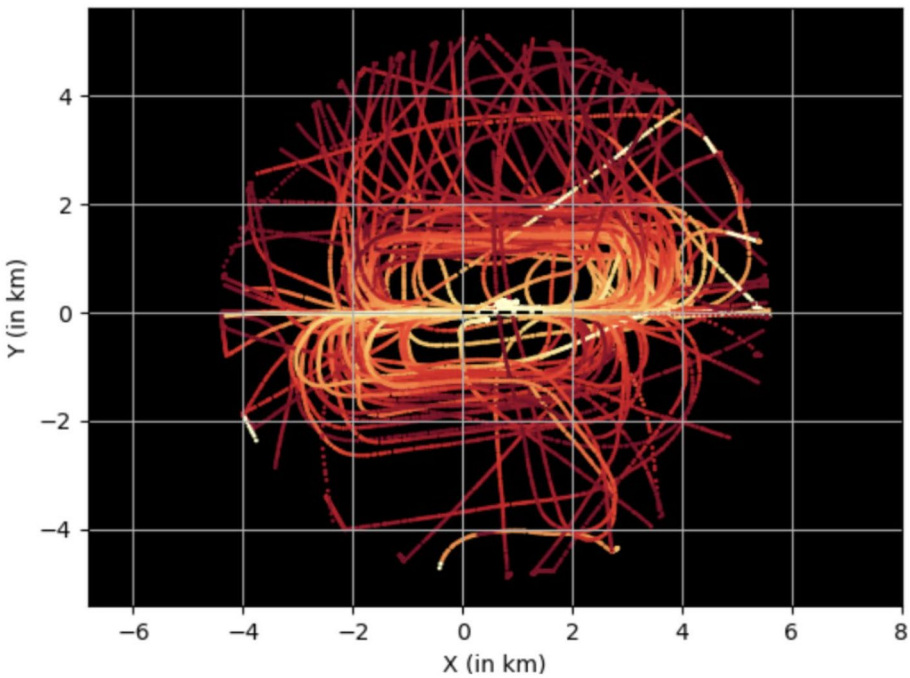


Fig. 6. Airport trajectory data visualization[2].

Subset	Sample Count (Number of Scenarios)		Total Samples	Total Trajectory Points
	Training Set	Test Set		
7days1	233	111	344	360,722
7days2	155	48	203	161,117
7days3	115	62	177	144,145
7days4	141	62	203	188,304

Table 1. TrajAir dataset statistics.

Average Displacement Error
ADE measures the average Euclidean distance between all predicted points and the corresponding ground-truth points along the trajectory. It reflects the overall performance of the model across the entire prediction time span. The calculation formula is shown in Eq. (21):

$$\mathrm{ADE} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{t_f}\sum_{t=1}^{t_f}\left|p_{i_p}^{t} - p_{i_{GT}}^{t}\right| \tag{21}$$

Where n is the number of trajectory samples, $p_{i_p}^{t}$ is the predicted position of the i-th trajectory at time t, and $p_{i_{GT}}^{t}$ is the corresponding ground-truth position.

Final displacement error
FDE represents the Euclidean distance between the predicted endpoint and the ground-truth endpoint of the trajectory. It directly measures the accuracy of the predicted final position. The calculation formula is shown in Eq. (22):

$$\mathrm{FDE} = \frac{1}{n}\sum_{i=1}^{n}\left|p_{i_p}^{t_f} - p_{i_{GT}}^{t_f}\right| \tag{22}$$

Where $p_{i_p}^{t_f}$ is the predicted position of the i-th trajectory at the final time step t_f, and $p_{i_{GT}}^{t_f}$ is the corresponding ground-truth position.

Maximum displacement error
MDE measures the maximum Euclidean distance between predicted and ground-truth positions over the entire prediction duration. It reflects the model's stability and the worst-case prediction error. The calculation formula is shown in Eq. (23):

$$\mathrm{MDE} = \frac{1}{n}\sum_{i=1}^{n}\max_{t=1,2,\ldots,t_f}\left|p_{i_p}^{t} - p_{i_{GT}}^{t}\right| \tag{23}$$

Smaller values for these three metrics indicate that the predicted trajectory is closer to the actual trajectory, implying better model performance and accuracy.

Experimental setup
In experiments, we utilize four subsets of the TrajAir dataset:7days1, 7days2, 7days3, and 7days4. We follow the original dataset configuration to divide the training and testing sets, as shown in Table. 2. We conducted experiments using batch training, where each training batch corresponds to a scenario sample. The minimum training unit is automatically divided based on the length of the aircraft trajectory in the scenario and the set trajectory window size. The number of training units in the datasets 7days1, 7days2, 7days3, and 7days4 are 46,895, 51,504, 48,767, and 59,578, respectively. The historical observation window size is set to 11, and the prediction window size is set to 120. The training runs for 50 epochs, using the Adam optimizer for backpropagation to train the network. The learning rate is set to 1e-4.

In the experiments, we apply three types of altitude feature extractors, and the parameter settings for these extractors are shown in Table. 2. Other network modules follow the default parameter settings in TrajAirNet.

Results and discussion
To validate the effectiveness of the proposed altitude-aware trajectory prediction algorithm, we conducted the following experiments.

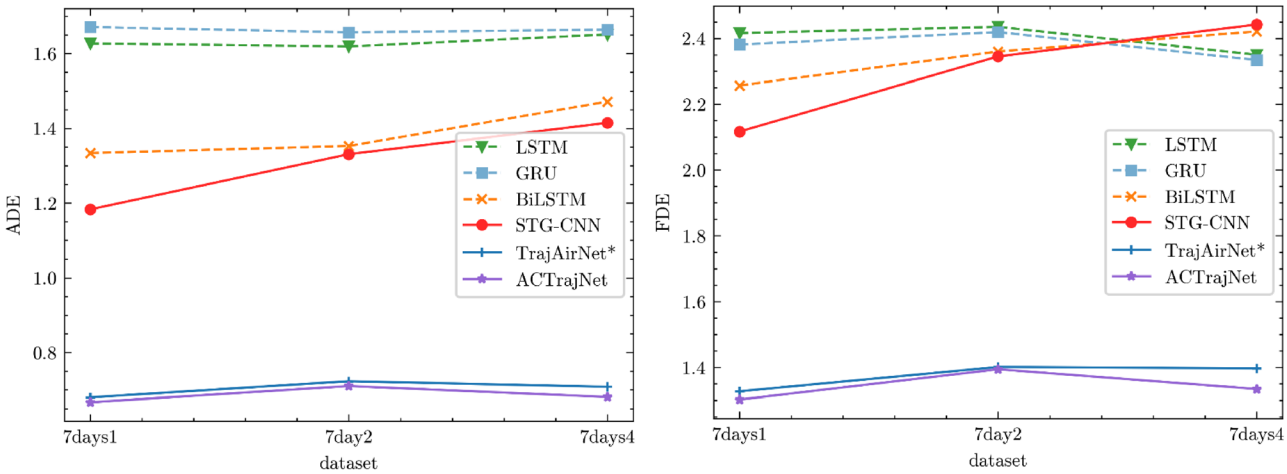
Comparison experiment of different network models
We compare the proposed ACTrajNet with several other network models, including LSTM, GRU, BiLSTM, STG-CNN[32], and TrajAirNet[2]. Since we do not consider interactions between targets, we have removed the graph

Feature Extractor	Hyperparameter	Parameter Description	Parameter Setting
TCN	tcn_channel_size	Size of the channels in TCN	256
	tcn_layers	Number of layers in TCN	2
	tcn_kernels	Kernel size of the convolution in TCN	4
	dropout	Dropout rate in TCN	0.05
BiLSTM	bilstm_hidden_size	Dimension of each hidden state in BiLSTM	256
	bilstm_layers	Number of layers in BiLSTM	4
	dropout	Dropout rate in BiLSTM	0.05
LSTM	lstm_hidden_size	Dimension of each hidden state in LSTM	256
	lstm_layers	Number of layers in LSTM	2

Table 2. Hyperparameters of height feature extractors.

	7days1			7days2			7days3			7days4		
Model Name	**ADE**	**FDE**	**MDE**	**ADE**	**FDE**	**MDE**	**ADE**	**FDE**	**MDE**	**ADE**	**FDE**	**MDE**
LSTM	1.627	2.416	3.162	1.619	2.435	2.745	1.670	3.292	3.831	1.651	2.350	3.562
GRU	1.671	2.381	2.420	1.657	2.419	2.759	1.671	3.213	3.359	1.664	2.334	3.413
BiLSTM	1.334	2.257	2.239	1.353	2.360	2.532	1.495	2.469	2.772	1.472	2.421	2.534
STG-CNN[32]	1.183	2.117	2.225	1.331	2.345	2.361	1.437	2.384	2.234	1.415	2.442	2.452
TrajAirNet[2]	0.680	1.327	**1.314**	0.723	1.401	1.469	0.841	1.613	1.698	0.708	1.397	1.454
ACTrajNet (Proposed Model)	**0.667**	**1.302**	1.360	**0.710**	**1.394**	**1.453**	**0.821**	**1.568**	**1.663**	**0.681**	**1.334**	**1.432**

Table 3. Comparison of different network models.



(a) ADE of different models on selected datasets

(b) FDE of different models on selected datasets

Fig. 7. Errors of different models on selected datasets.

attention network module from TrajAirNet. The network layers for LSTM, GRU, and BiLSTM are set to 4 layers, 4 layers, and 8 layers, respectively, with a hidden state dimension of 512 for each layer. The hyperparameters for STG-CNN and TrajAirNet are set according to the original papers. The final experimental results are shown in Table. 3. Bolded text in the table indicates the smallest value for each evaluation metric, while underlined text indicates the second smallest. For all three evaluation metrics, smaller values correspond to better prediction performance. Fig. 7(a) shows the ADE comparison between ACTrajNet and other models on a subset of the dataset, while Fig. 7(b) presents the FDE comparison between ACTrajNet and other models on a subset of the dataset.

As shown in Table. 3, Fig. 7(a) and Fig. 7(b) , ACTrajNet significantly outperforms the baseline model in terms of ADE and FDE. On the one hand, this is because ACTrajNet independently extracts altitude features through TCN and dynamically allocates weights using channel attention, enabling the model to focus on key channels sensitive to altitude variations. This mechanism effectively enhances the model's ability to model frequent ascent and descent trajectories, thereby reducing the overall ADE and FDE in trajectory prediction.

On the other hand, the decoder uses CVAE, which generates diverse predicted trajectories by sampling from the latent space[33]. This probabilistic generation approach better captures the randomness of aircraft trajectories in non-towered airspace, thus covering the true trajectory distribution in most scenarios and optimizing the ADE and FDE metrics.

However, ACTrajNet shows slightly suboptimal performance on the MDE metric due to two main factors. In feature extraction, although the TCN module effectively captures the overall trend of time-series data, its low-pass filtering property weakens the representation of sudden altitude changes. As a result, it struggles to accurately capture key displacement mutations. In the generation mechanism, the random sampling process of CVAE enhances prediction diversity under normal conditions. However, during extreme altitude fluctuations, it may generate unrealistic trajectory predictions that exceed actual flight physics constraints, significantly impacting the maximum error metric. The combined effect of these factors ultimately leads to an increase in the MDE metric.

To better understand how the CVAE module performs under extreme altitude changes, we conducted a detailed analysis on trajectory segments where the altitude change rate exceeds 40 meters per second. We found that the CVAE's random sampling mechanism can occasionally produce trajectory points that briefly violate physical constraints. However, these anomalies usually last for a short time, and the model demonstrates strong self-correction ability in subsequent prediction steps. This helps explain why ACTrajNet performs exceptionally well on key metrics like ADE and FDE, while only showing minor effects on the MDE metric.

Future work will explore the integration of a deformable convolutional network in feature extraction. By dynamically adjusting the convolution kernel shape, the model can enhance its ability to capture abrupt trajectory changes. In the generation process, flight dynamics constraints will be embedded into the decoding stage to guide the sampling process, ensuring compliance with real-world flight patterns. This approach aims to reduce the MDE metric.

Comparison of altitude prediction performance

To further verify ACTrajNet's ability to capture dynamic changes in the vertical direction, we used the same model hyperparameter settings as in the comparison experiments with other models. We calculated the prediction error metrics for the altitude component: Altitude ADE (AADE), Altitude FDE (AFDE), and Altitude MDE (AMDE). Unlike ADE, FDE, and MDE, these metrics focus specifically on evaluating prediction errors in the altitude dimension. The final experimental results are shown in Table. 4. Fig. 8(a) presents the comparison of AADE among different models on the dataset, Fig. 8(b) shows the comparison of AFDE, and Fig. 8(c) shows the comparison of AMDE.

As shown in Table. 4, Fig. 8(a) , Fig. 8(b) , and Fig. 8(c) , ACTrajNet performs significantly better than all baseline models in terms of AADE and AFDE. This clearly demonstrates that independently extracting altitude features and dynamically fusing them using a channel attention mechanism greatly enhances the model's ability to capture frequent vertical fluctuations. This strategy is a key factor contributing to the overall performance improvement of ACTrajNet. Although the AMDE metric is somewhat limited by the inherent low-pass filtering nature of the TCN convolution kernel–reducing the model's ability to represent sharp altitude changes, and the CVAE's random sampling mechanism may further amplify prediction uncertainty when the model underfits abrupt variations (leading to occasional instantaneous outliers), ACTrajNet still achieves the best performance on this metric.

Ablation study on altitude feature fusion module

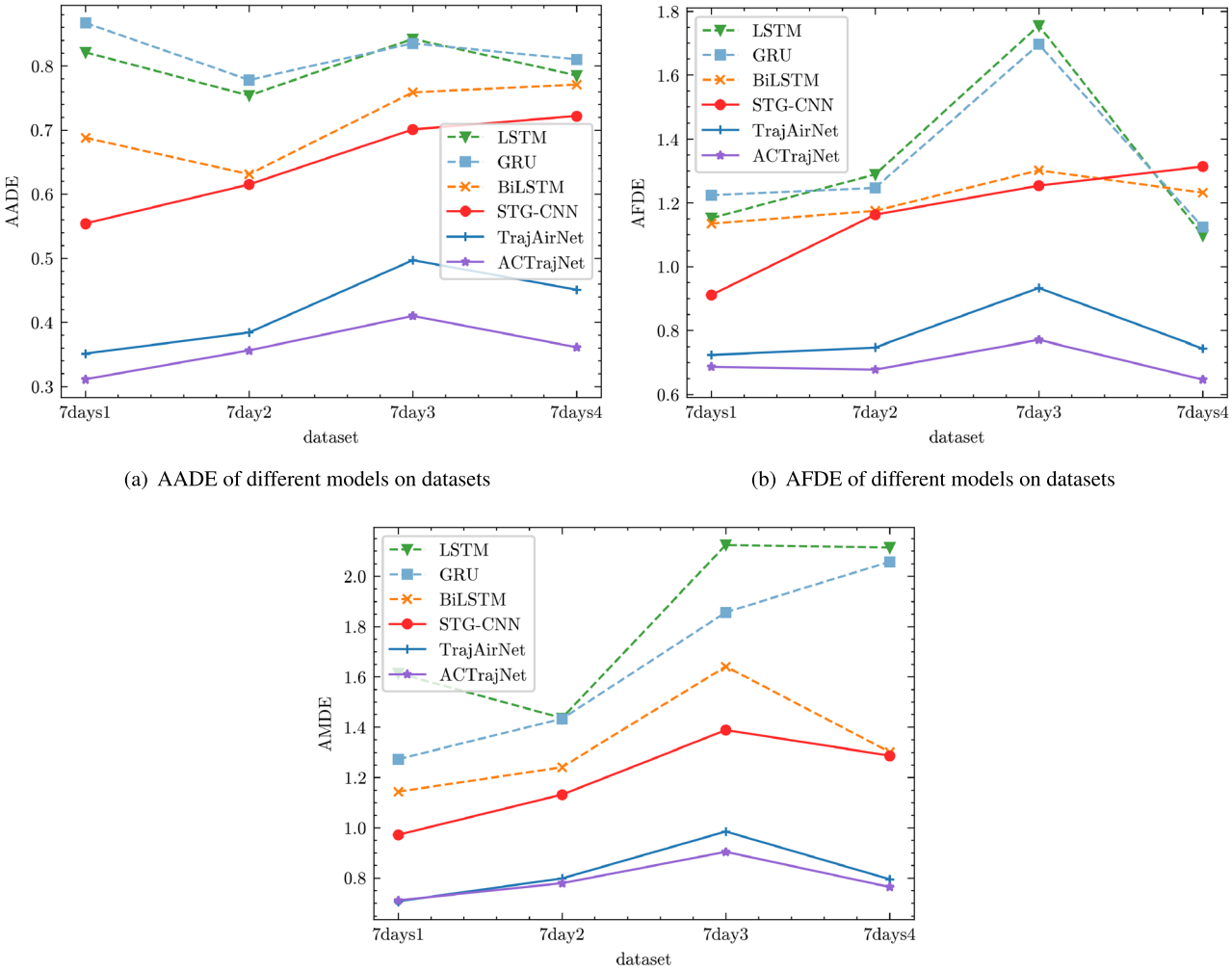
We compare three fusion encoding methods on four datasets from TrajAir. To maximize the benefit of the fusion encoding, we implemented and compared three altitude feature extractors: LSTM, BiLSTM, and TCN. The experimental results are shown in Table. 5. The symbol "w/" in the table indicates that the altitude fusion module was added to the proposed model, the baseline is TrajAirNet. Fig. 9(a) compares the ADE of different fusion mechanisms and feature extractors, while Fig. 9(b) compares their FDE.

From Table. 5, Fig. 9(a) and Fig. 9(b) , we observe that the introduction of the three altitude feature fusion module effectively improves the model performance. In particular, the combination of the TCN and CAF as the final altitude information fusion encoding scheme shows the most significant improvement. Across the four datasets, the ADE, FDE, and MDE metrics improve by an average of 2.47%, 2.42%, and 0.29%, respectively, compared to the baseline without fusion encoding.

In comparison, the VCC method demonstrates inconsistent performance improvements and even underperforms the baseline model on certain datasets (7days1, 7days2). The primary reason is that altitude

	7days1			7days2			7days3			7days4		
Model Name	**AADE**	**AFDE**	**AMDE**	**AADE**	**AFDE**	**AMDE**	**AADE**	**AFDE**	**AMDE**	**AADE**	**AFDE**	**AMDE**
LSTM	0.821	1.152	1.614	0.754	1.289	1.436	0.842	1.755	2.124	0.785	1.098	2.114
GRU	0.867	1.224	1.273	0.778	1.247	1.433	0.835	1.696	1.857	0.810	1.124	2.057
BiLSTM	0.688	1.135	1.144	0.631	1.175	1.241	0.759	1.302	1.641	0.771	1.232	1.301
STG-CNN	0.554	0.911	0.972	0.615	1.163	1.132	0.701	1.254	1.388	0.722	1.314	1.287
TrajAirNet	0.351	0.724	**0.707**	0.384	0.747	0.798	0.497	0.933	0.985	0.451	0.744	0.795
ACTrajNet	**0.311**	**0.687**	0.711	**0.356**	**0.678**	**0.779**	**0.410**	**0.771**	**0.904**	**0.361**	**0.647**	**0.764**

Table 4. Comparison of altitude prediction errors across different models.



(a) AADE of different models on datasets

(b) AFDE of different models on datasets

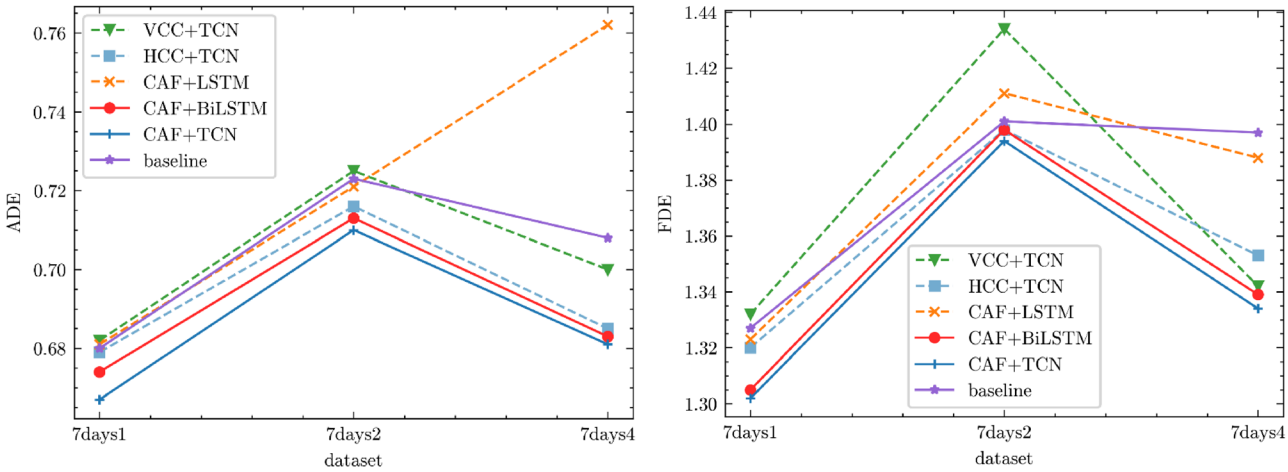
(c) AMDE of different models on datasets

Fig. 8. Errors of different models on datasets.

Fusion Mechanism	Altitude Feature Extractor	7days1			7days2			7days3			7days4		
		ADE	FDE	MDE	ADE	FDE	MDE	ADE	FDE	MDE	ADE	FDE	MDE
w/ VCC	LSTM	0.722	1.413	1.481	0.735	1.452	1.503	0.875	1.652	1.683	0.715	1.410	1.465
	BiLSTM	0.693	1.328	1.331	0.732	1.443	1.491	0.855	1.611	1.643	0.703	1.357	1.432
	TCN	0.682	1.332	1.336	0.725	1.434	1.485	0.828	1.565	1.666	0.700	1.342	1.439
w/ HCC	LSTM	0.708	1.345	1.345	0.726	1.416	1.483	0.833	1.575	1.672	0.734	1.431	1.498
	BiLSTM	0.685	1.328	1.311	0.720	1.408	1.461	0.827	1.566	1.653	0.712	1.399	1.415
	TCN	0.679	1.320	1.317	0.716	1.398	1.469	0.823	1.574	1.648	0.685	1.353	1.463
w/ CAF	LSTM	0.681	1.323	1.312	0.721	1.411	1.462	0.826	1.572	1.653	0.762	1.388	1.458
	BiLSTM	0.674	1.305	**1.300**	0.713	1.398	**1.425**	0.823	**1.562**	**1.633**	0.683	1.339	**1.394**
	TCN	**0.667**	**1.302**	1.360	**0.710**	**1.394**	1.453	**0.821**	1.568	1.663	**0.681**	**1.334**	1.432
baseline	\	0.680	1.327	1.314	0.723	1.401	1.469	0.841	1.613	1.698	0.708	1.397	1.454

Table 5. Ablation study results of the altitude feature fusion module.

features are directly concatenated with flattened trajectory representations along the vertical axis. This simple feature-level fusion fails to establish deep interactions within the model. Additionally, since altitude features and trajectory representations remain independent after concatenation, the CVAE module struggles to capture their nonlinear correlations. This leads to an imbalanced distribution of information, making it difficult for the model



(a) ADE of different fusion mechanisms and feature extractors

(b) FDE of different fusion mechanisms and feature extractors

Fig. 9. Errors of different fusion mechanisms and feature extractors.

to effectively integrate altitude semantics during parameter updates, ultimately impacting overall prediction performance.

HCC improves upon VCC by concatenating altitude features horizontally at the end of each channel. This allows each channel to incorporate vertical information, followed by a fully connected layer for dimensional alignment and nonlinear activation. While this method facilitates better local information fusion compared to VCC, its fixed concatenation strategy lacks adaptability to varying levels of altitude importance across different flight scenarios.

CAF, on the other hand, introduces a channel attention mechanism, enabling dynamic weight allocation within the model. Specifically, CAF first maps the altitude features extracted by the TCN to the same channel dimension as the high-dimensional trajectory representation using a fully connected layer. It then applies a sigmoid function to generate attention weights, which are multiplied element-wise with the channel features. This process not only strengthens the deep coupling between altitude and trajectory information but also allows the network to dynamically adjust each channel's contribution based on flight state changes. As a result, CAF significantly enhances sensitivity to frequent vertical fluctuations and improves feature capture, providing more discriminative input for the subsequent CVAE module, ultimately leading to better overall prediction performance.

Based on the above analysis, we ultimately selects CAF as the final altitude feature fusion encoding method. Additionally, experimental results on the 7days3 dataset, as shown in Tables 3 and 5, indicate relatively poorer prediction performance. This reflects the challenges of trajectory prediction in tower-free terminal airspace, primarily due to the diversity of aircraft types and the flexibility of operational patterns. Nevertheless, the proposed model maintains relatively stable performance overall, further demonstrating the effectiveness of the introduced channel attention fusion mechanism.

Hyperparameter sensitivity analysis experiment

To investigate the sensitivity of the proposed TCN-based altitude feature extractor to key hyperparameters, we conducted ablation experiments within the ACTrajNet (CAF fusion) framework by fixing other settings and varying the TCN channel size, number of layers, and kernel size. The results are shown in Table. 6.

According to Table. 6, as the number of channels increases from 32 to 256, both ADE and FDE improve steadily, indicating that a larger channel capacity enhances the model's feature representation ability. However, when the channel size is further increased to 512, performance saturates or even declines, suggesting that over-parameterization may introduce redundant noise. A 2-layer TCN achieves the best results on most datasets–using only 1 layer limits the receptive field, making it difficult to capture long-term dependencies, while using more than 3 layers causes performance fluctuations due to gradient vanishing and overfitting. A kernel size of 4 yields the best overall performance: smaller kernels fail to capture abrupt local altitude changes, while larger kernels tend to overly smooth the features and lose important details.

Combining the effects of all parameter adjustments, we find that the TCN achieves optimal performance with 256 channels, 2 layers, and a kernel size of 4.

Validation of model generalization

Publicly available trajectory datasets for non-towered terminal airspace are quite limited, with TrajAir being one of the few. To further evaluate the generalization ability of the ACTrajNet model across different flight scenarios, we select flight trajectory samples from tower-controlled ADS-B data that exhibit significant vertical fluctuations. These samples are used to simulate situations with frequent altitude changes, thereby helping to assess the model's generalization performance. As shown in Fig. 10.

Hyperparameter	Value	7days1			7days2			7days3			7days4		
		ADE	FDE	MDE	ADE	FDE	MDE	ADE	FDE	MDE	ADE	FDE	MDE
tcn_channel_size	32	0.689	1.329	1.385	0.731	1.418	1.481	0.846	1.579	1.688	0.699	1.352	1.451
	64	0.685	1.317	1.387	0.738	1.405	1.466	0.841	1.581	1.678	0.689	1.355	1.447
	128	0.672	1.311	**1.358**	0.715	1.406	1.459	0.830	1.574	**1.661**	0.689	1.341	1.439
	256	**0.667**	**1.302**	1.360	**0.710**	**1.394**	**1.453**	**0.821**	**1.568**	1.663	**0.681**	**1.334**	**1.432**
	512	0.675	1.314	1.362	0.721	1.411	1.462	0.835	1.579	1.667	0.691	1.347	1.445
tcn_layers	1	0.679	1.321	1.385	0.716	1.411	1.475	0.829	1.581	1.679	0.699	1.354	1.443
	2	**0.667**	1.302	**1.360**	**0.710**	**1.394**	**1.453**	0.821	**1.568**	**1.663**	**0.681**	**1.334**	**1.432**
	3	0.669	**1.298**	1.363	0.725	1.401	1.458	**0.819**	1.572	1.668	0.685	1.339	1.438
	4	0.679	1.317	1.375	0.729	1.409	1.465	0.832	1.577	1.671	0.688	1.339	1.447
	5	0.681	1.315	1.378	0.726	1.415	1.472	0.827	1.576	1.680	0.693	1.351	1.450
tcn_kernels	2	0.671	1.321	1.376	0.724	1.414	1.473	0.838	1.588	1.674	0.702	1.351	1.449
	3	0.670	1.314	1.366	0.719	1.406	1.468	0.829	1.577	1.670	0.698	1.345	1.445
	4	**0.667**	**1.302**	**1.360**	**0.710**	**1.394**	**1.453**	**0.821**	**1.568**	**1.663**	**0.681**	**1.334**	**1.432**
	5	0.679	1.322	1.385	0.715	1.411	1.464	0.835	1.577	1.672	0.695	1.339	1.447
	6	0.677	1.316	1.374	0.722	1.416	1.471	0.839	1.581	1.679	0.705	1.355	1.457

Table 6. Sensitivity analysis of ACTrajNet hyperparameters.

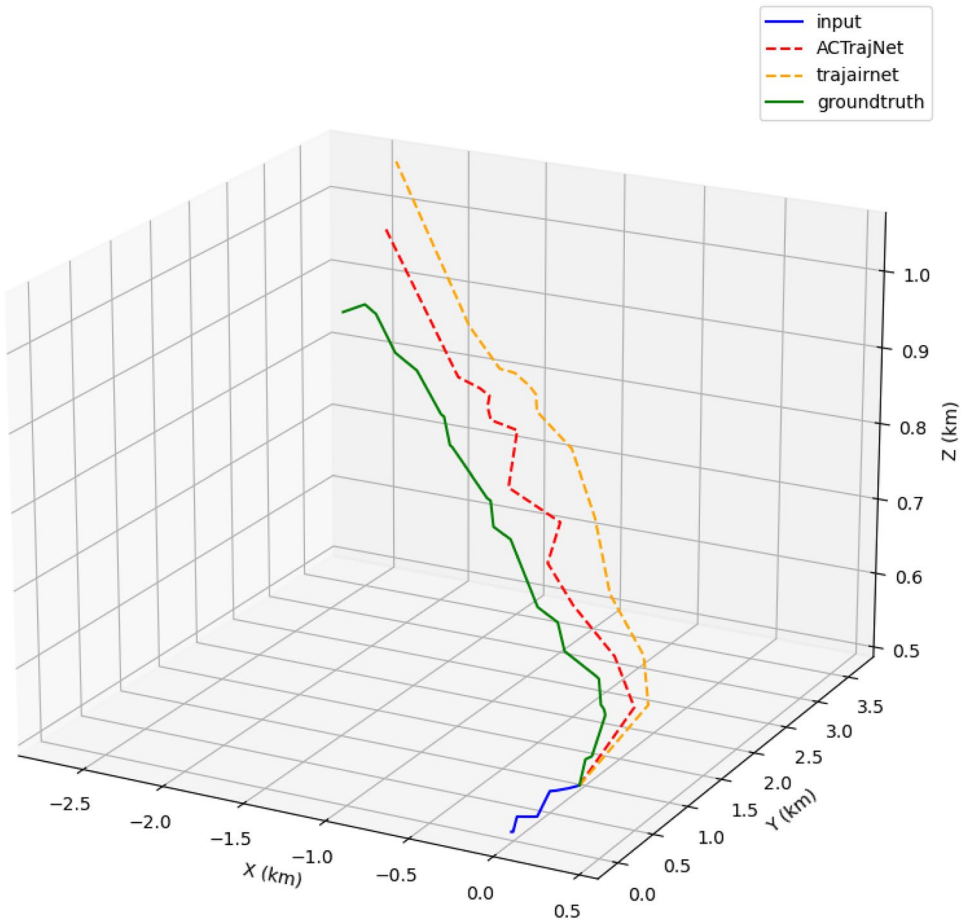


Fig. 10. ADS-B trajectory prediction visualization. The solid blue line represents the input historical trajectory, consisting of 11 past trajectory points. The solid green line shows the actual trajectory, made up of 120 real trajectory points. The red dashed line indicates the prediction results of our model ACTrajNet, consisting of 120 trajectory points. The yellow dashed line represents the prediction results of the baseline model TrajAirNet, also consisting of 120 trajectory points.

The experimental results show that, compared to the baseline model TrajAirNet, the trajectories predicted by ACTrajNet align more closely with the actual flight paths overall. While TrajAirNet captures the main trends in the horizontal direction, its predictions significantly underestimate the vertical fluctuations and fail to accurately reflect the sharp altitude changes. This highlights its limitations in dynamically modeling altitude features.

However, it should be noted that due to the inherent characteristics of tower-controlled ADS-B trajectory data, the prediction performance of ACTrajNet on this type of dataset still has certain limitations and has not yet reached the ideal level achieved on the TrajAir dataset.

ACTrajNet not only effectively improves performance on the TrajAir dataset but also demonstrates a certain degree of generalization ability, adapting well to trajectory prediction tasks with significant vertical fluctuations. This validates the effectiveness of the model design in addressing the challenges of complex flight scenarios in non-towered terminal airspace.

Conclusion

This paper presents ACTrajNet, an altitude-aware trajectory prediction model for non-towered terminal airspace. Experiments show that ACTrajNet reduces ADE and FDE on the TrajAir dataset by 2.47% and 2.42% through independent altitude feature extraction and dynamic channel attention fusion. Furthermore, experimental results demonstrate that TCN combined with CAF achieves the best performance in altitude feature fusion for trajectory prediction. Future work will further explore aircraft trajectory prediction in multi-aircraft interaction scenarios and in complex airspace.

Data availability

The dataset used in the current study is open-source and can be downloaded from https://theairlab.org/trajair/. Code is available at this repository: https://github.com/zhixiangbaipiao/ACTrajNet.

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Author contributions
H.Z.: Conceptualization, Methodology; J.H.: Data analysis and Writing; Q.T.: Validation, Reviewing and Editing;X.L.:Investigation and Writing. S.H.:Revising the manuscript based on review comments.All authors reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
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